List and Bag, π and $\sqrt{\pi}$

The missing ordinary row of the decategorification table, and a Gaussian functional that counts S_n-quotients

Abstract

The ordinary generating series $O(x) = \sum x^n$ and the exponential one $E(x) = \sum x^n/n!$ satisfy

$$\int_{\mathbb{R}} O(-x^2)\,dx = \pi, \qquad \int_{\mathbb{R}} E(-x^2)\,dx = \sqrt{\pi}.$$

Both are $\Gamma(\frac{1}{2})$ to a power. We read this containerwise. O and E are not two generating functions of one species but the exponential generating functions of two species, L (linear orders) and E (sets), which as containers are List and Bag: the same shapes and positions, with and without the S_n-quotient. The functional $F \mapsto \int_{\mathbb{R}} F(-x^2)\,dx$ sees the quotient, and Theorem 5.1 says exactly what it charges for it: dividing the coefficients by $n!$ once costs one factor of $\Gamma(\frac{1}{2})$. The exponent 2 in the π case and the $-s$ in the general case have different origins — the first is the Gamma reflection formula at the self-dual Mellin exponent $\frac{1}{2}$, which is what the squaring in $-x^2$ is for, and the second is the $n!$-divisions. Two obstructions are recorded honestly: **Poly** has no additive inverses, so $-x^2$ is not an object and the computation lives in the Grothendieck ring of virtual species; and $\int_{\mathbb{R}}$ is not a container operation but a second decategorification, whose nearest precedent is Spivak's entropy of a polynomial functor.

1 The observation

Proposition 1.1. $\displaystyle\int_{\mathbb{R}} \frac{dx}{1+x^2} = \pi$ *and* $\displaystyle\int_{\mathbb{R}} e^{-x^2}\,dx = \sqrt{\pi}$.

Both are elementary; $O(-x^2) = 1/(1+x^2)$ because $O(x) = 1/(1-x)$, and $E(-x^2) = e^{-x^2}$ because $E(x) = e^x$. Checked numerically to 30 digits: **verified**. The integral must be taken over all of $\mathbb{R}$; over $[0,\infty)$ one gets $\pi/2$ and $\sqrt{\pi}/2$ and the pattern is obscured.

Remark 1.2 (the termwise reading fails, and that is the content). One cannot obtain either number by integrating the series. $\int_{\mathbb{R}} x^{2n}\,dx = \infty$ for every n, so no rearrangement is available. Worse, for O the series $\sum(-x^2)^n$ converges only for $|x| < 1$ while $1/(1+x^2)$ is integrable on all of $\mathbb{R}$: the functional is applied to the analytic continuation, not to the series. The generating function is doing work here beyond bookkeeping, which is why the identity is a statement about the *middle rung* of the decategorification diagram and not about coefficients.

2 The missing row

Day convolution along a monoidal index category $(\mathcal{A}, \otimes)$ is the unique monoidal structure making the Yoneda embedding strong monoidal, and $\mathcal{A}$ is an input. Varying it varies the decategorified product. The table of `q-dirichlet.tex` §2 reads:

$\mathcal{A}$	Day convolution	decategorification
$(\mathbf{FinSet}^{\mathrm{iso}}, \sqcup)$	species product	$\sum a_n x^n/n!$ (Cauchy/EGF)
$(\mathbb{N}_{>0}, \times)$	Dirichlet convolution	$\sum a_n n^{-s}$
$(\mathbf{FinSet}^{\mathrm{iso}}_{\neq\emptyset}, \times)$	arithmetic product	$\sum a_n n^{-s}/n!$

The row the present question is about is not there. Let $\mathbb{N}^\delta$ be the *discrete* monoid $(\mathbb{N}, +)$, that is $\mathbf{FinSet}^{\mathrm{iso}}$ with every non-identity automorphism deleted.

$\mathcal{A}$	Day convolution	decategorification
$(\mathbb{N}^\delta, +)$	Cauchy product of **Set**-sequences	$\sum a_n x^n$ (OGF)

Proposition 2.1 (*ours*)**.** *The two rows* $(\mathbb{N}^\delta, +)$ *and* $(\mathbf{FinSet}^{\mathrm{iso}}, \sqcup)$ *differ only in whether the index category remembers* $\mathrm{Aut}([n]) = S_n$*, and the two decategorifications differ only by the divisor* $n! = |S_n|$*. Hence* O *and* E *of Proposition 1.1 are one sequence of shapes,* $a_n = 1$*, decategorified along two index categories related by the inclusion* $\mathbb{N}^\delta \hookrightarrow \mathbf{FinSet}^{\mathrm{iso}}$*.*

This is the correct statement but it is not the most useful one, because it makes the difference live in the *functor* $\mathcal{A} \mapsto$ (series ring) rather than in the objects. The next section moves the difference into the containers, where it can be seen.

3 O and E are List and Bag

Recall the two species of Joyal [3], in the notation of Bergeron–Labelle–Leroux [1]:

$$L[n] = \{\text{linear orders on } [n]\}, \quad |L[n]| = n!, \qquad E[n] = \{*\}, \quad |E[n]| = 1.$$

Their exponential generating functions are

$$\sum_{n\geq 0} |L[n]| \frac{x^n}{n!} = \sum_{n\geq 0} x^n = \frac{1}{1-x} = O(x), \qquad \sum_{n\geq 0} |E[n]| \frac{x^n}{n!} = e^x = E(x).$$

Proposition 3.1 (*ours*, and immediate)**.** *O and E are both exponential generating functions. O is the EGF of the species of linear orders; E is the EGF of the species of sets. The "ordinary versus exponential" distinction of §2 is the same distinction as "L versus E", seen from the other side.*

Containerwise, with $(S \rhd P)$ denoting shapes S and positions P:

$$\mathrm{List} = \big(\mathbb{N} \rhd n \mapsto [n]\big), \qquad [\![\mathrm{List}]\!]X = \sum_{n\geq 0} X^n,$$

the free monoid, and Bag is the same container with the S_n-action on positions divided out,

$$[\![\mathrm{Bag}]\!]X = \sum_{n\geq 0} X^n/S_n,$$

a *quotient container* in the sense of Kock [5]: a polynomial functor over groupoids rather than over sets. So:

container	species	EGF	$\int_{\mathbb{R}} \cdot(-x^2)$
List	L, linear orders	$1/(1-x)$	$\pi = \Gamma(\frac{1}{2})^2$
Bag	E, sets	e^x	$\sqrt{\pi} = \Gamma(\frac{1}{2})^1$

This is the row of `q-container.tex` §"Related work" that names Kock's species/quotient-container bridge as "the structural model to imitate". The present note is a small instance of that bridge in which the imitation can be carried out completely, because both sides are one-dimensional.

4 Borel is the de-quotienting map

The reason the two containers are comparable at all is that List's series is the Laplace transform of Bag's. For $\operatorname{Re} x < 1$,

$$O(x) \;=\; \frac{1}{1-x} \;=\; \int_0^\infty e^{-t} e^{xt}\, dt \;=\; \int_0^\infty e^{-t} E(xt)\, dt, \tag{1}$$

and the mechanism is coefficientwise: $\int_0^\infty e^{-t} t^n\, dt \;=\; n! \;=\; |L[n]|$. The Gamma integral is precisely what restores the S_n that Bag divided out.

Proposition 4.1 (*ours*)**.** *Substituting $x \mapsto -x^2$ in (1) and exchanging the order of integration gives*

$$\int_{\mathbb{R}} O(-x^2)\, dx = \int_0^\infty e^{-t} \Big(\int_{\mathbb{R}} e^{-tx^2}\, dx \Big) dt = \int_0^\infty e^{-t} \sqrt{\pi/t}\; dt = \sqrt{\pi}\, \Gamma(\tfrac{1}{2}) = \Gamma(\tfrac{1}{2})^2.$$

The two factors of $\Gamma(\frac{1}{2})$ have visibly different sources: the inner Gaussian supplies one, and the $t^{-1/2}$ left behind by the Gaussian scaling $x \mapsto x/\sqrt{t}$ is eaten by the Borel integral to supply the other. So the π of the List row is the $\sqrt{\pi}$ of the Bag row multiplied by the cost of de-quotienting.

5 The theorem

Proposition 4.1 is one step of a family. Let

$$F_s(z) \;=\; \sum_{n \geq 0} \frac{z^n}{\Gamma(n+1)^s} \;=\; \sum_{n \geq 0} \frac{z^n}{(n!)^s}, \qquad s \in \mathbb{Z}_{\geq 0},$$

so that $F_0 = O$, $F_1 = E$, and $F_2(z) = {}_0F_1(;1;z)$ with $F_2(-x^2) = J_0(2x)$. Dividing by $n!$ once more is one more S_n-quotient taken.

Theorem 5.1 (*ours*)**.** *For $m \geq 2$ and $s \geq 0$,*

$$\int_0^\infty F_s(-x^m)\, dx \;=\; \frac{\pi}{m \sin(\pi/m)} \cdot \Gamma\big(1 - \tfrac{1}{m}\big)^{-s},$$

in the sense of the Mellin regularisation below. In particular at $m = 2$,

$$\boxed{\int_{\mathbb{R}} F_s(-x^2)\, dx \;=\; \Gamma(\tfrac{1}{2})\, \Gamma(\tfrac{1}{2}) \cdot \Gamma(\tfrac{1}{2})^{-s} \;=\; \Gamma(\tfrac{1}{2})^{\,2-s} \;=\; \pi^{(2-s)/2}.}$$

Proof. Ramanujan's master theorem in the form $\int_0^\infty x^{\sigma-1} \sum_{k \geq 0} (-1)^k \varphi(k) x^k\, dx = \Gamma(\sigma)\Gamma(1-\sigma)\varphi(-\sigma)$, valid for φ analytic in a right half-plane of suitable growth and $0 < \operatorname{Re}\sigma < 1$. Substitute $u = x^m$, so $dx = \frac{1}{m} u^{1/m - 1} du$:

$$\int_0^\infty F_s(-x^m)\, dx = \frac{1}{m} \int_0^\infty u^{\sigma-1} F_s(-u)\, du, \qquad \sigma = \tfrac{1}{m} \in (0,1).$$

Here $\varphi(k) = \Gamma(k+1)^{-s}$, which is entire since $1/\Gamma$ is entire, so $\varphi(-\sigma) = \Gamma(1 - \frac{1}{m})^{-s}$. The reflection formula gives $\Gamma(\sigma)\Gamma(1-\sigma) = \pi/\sin(\pi\sigma)$, and the stated formula follows. At $m = 2$ we have $\sigma = \frac{1}{2}$, $\pi/\sin(\pi/2) = \pi = \Gamma(\frac{1}{2})^2$, and $\varphi(-\frac{1}{2}) = \Gamma(\frac{1}{2})^{-s}$; doubling for the two-sided integral cancels the $1/m = \frac{1}{2}$. □

Remark 5.2 (what is special about $m = 2$, i.e. about the *pair*). The two Γ-factors in Theorem 5.1 are $\Gamma(\frac{1}{m})\Gamma(1-\frac{1}{m})$ from the reflection formula and $\Gamma(1-\frac{1}{m})^{-s}$ from the $n!$-divisions. At $m = 2$, and only there, $\frac{1}{m} = 1 - \frac{1}{m}$: the self-dual point of $\sigma \mapsto 1 - \sigma$. That is the unique exponent at which the whole answer is a power of a single number rather than a product of two unrelated ones — and hence the unique one at which the pattern "π, then $\sqrt{\pi}$" is visible at all. Containerwise, x^2 is $y \otimes y$, the *pair*: substituting the pair container is what places the Mellin exponent at the self-dual point.

Remark 5.3 (verification, and where convergence stops). Numerically (`mpmath`, 25–30 digits): $s = 0$ gives π **verified**; $s = 1$ gives $\sqrt{\pi}$ **verified**; $s = 2$ gives $1 = \Gamma(\frac{1}{2})^0$ **verified**, as $\int_{\mathbb{R}} J_0(2x)\,dx = 1$, Abel-convergent and confirmed by damping $e^{-\varepsilon x}$ with $\varepsilon = 2^{-1}, \ldots, 2^{-5}$ $(0.9701, 0.9923, 0.9981, 0.9995, 0.9999)$. For $m = 3$, $s = 1$ the formula reduces to $\Gamma(4/3) = \Gamma(1/3)/3$, confirmed symbolically by the reflection formula **verified**.

For $s \geq 3$ the integral is *not* absolutely convergent and need not be conditionally convergent either: ${}_0F_{s-1}(;1,\ldots,1;-x^2)$ oscillates in $x^{2/s}$, so it is not periodic and ordinary oscillatory quadrature does not apply. The theorem's claim for $s \geq 3$ is a claim about the Mellin/Abel-regularised value, which is what Ramanujan's master theorem computes. This is the same caveat as Remark 1.2 one level up: the identity is about the analytic object, not about any convergent sum or integral of pieces.

Corollary 5.4 (the reading)**.** *Write* $\Gamma(\frac{1}{2})^{2-s}$ *as* $\Gamma(\frac{1}{2})^2 \cdot \Gamma(\frac{1}{2})^{-s}$. *The* 2 *is geometry — the reflection formula at the self-dual exponent forced by the pair container — and the* $-s$ *is combinatorics: each* S_n*-quotient taken costs exactly one factor of* $\Gamma(\frac{1}{2})$. *The* List $\to$ Bag *step of §3 is* $s = 0 \to 1$.

This is the kind of statement the programme box of `q-container.tex` §"The programme" asks for: a coefficient that looks arbitrary at the analytic level (π here, $\sqrt{\pi}$ there) forced at the container level, by counting quotients.

6 Two obstructions

Neither of the following is a gap in Theorem 5.1, which is a theorem about series. Both are gaps in the attempt to make the *whole* statement live at the container level, and both should be recorded before the reading of Corollary 5.4 is believed to be more than suggestive.

6.1 Poly has no additive inverses, so $-x^2$ is not an object

Poly, the category of polynomial endofunctors on **Set** under $\sqcup$ and $\otimes$, is a rig category: there is no additive inverse of $y \otimes y$, and $-x^2$ names no object. Likewise $\mathbf{Cont}_q$. The substitution $F \mapsto F(-x^2)$ is therefore not container composition $\triangleright$ but composition in the ring of *virtual* species [4, 7], that is in the Grothendieck ring K_0 of the species category, where formal differences exist.

Question 6.1. Does $\mathbf{Cont}_q$ have a usable K_0? Specifically: is the Grothendieck group of q-containers under $+$ (coproduct) a ring under the product of `q-container.tex` Def. (product of containers), and does the q-series map $f_C(x) = \sum_s x^{\dim Ps}$ extend to it as a ring homomorphism into $\mathbb{Z}[[x]]$?

Warning 6.2. This is the same shape of failure as the divergence section of `q-container.tex`: containers resisted Hall convolution because the construction needed is a convolution on a stack rather than a Day convolution, and they resist subtraction because **Poly** is a rig and not a ring. The two should not be assumed independent. In particular, do not take the alternating signs of $F_s(-x^2)$ as evidence of anything container-theoretic until Question 6.1 is settled.

6.2 $\int_{\mathbb{R}}$ is a second decategorification, not a container operation

The table of `q-container.tex` §"decategorification" lists one operation per rung and every one of them is algebraic: coproduct, product, Day convolution, Hall product, derivative ∂ (hyperplanes, decategorifying to the Jackson D_q), shift σ, Ind/Res. The species derivative is pointing — the one-hole context, the zipper — and it has a container meaning. Integration in the sense used here does not: $\int_{\mathbb{R}}$ is a measure-theoretic functional on the middle rung, not the formal antiderivative, and the composite

$$\text{container} \longrightarrow \text{series} \longrightarrow \text{number}$$

factors the decategorification into two stages of which only the first is structural.

The nearest precedent is Spivak. In [6] and [2] a Dirichlet polynomial is read as a set-theoretic bundle, hence as an empirical distribution, and assigned a Shannon entropy — a number extracted from a polynomial functor by a functional that is not one of **Poly**'s own operations, and which is nonetheless functorial where it needs to be. That is the same shape of move as the one made here, and it is the right citation for it.

Conjecture 6.3 (*ours*, unverified)**.** *There is a compatibility statement of the following form. Let $\mathcal{B}$ denote the Borel/Laplace transform of (1), viewed as the de-quotienting map from symmetric to rigid structures, and let $\Phi(F) = \int_{\mathbb{R}} F(-x^2)\,dx$. Then $\Phi \circ \mathcal{B} = \Gamma(\frac{1}{2}) \cdot \Phi$ on the relevant domain; Theorem 5.1 is the restriction of this to the family F_s.*

If Conjecture 6.3 is right it is more than the entropy story has: a functional that is *not* a container operation nevertheless intertwining with a map that *is* one, by a single scalar. That scalar is $\Gamma(\frac{1}{2})$, and the reason it is Γ is that Γ is where $n!$ lives.

7 Next

1. **Settle Question 6.1.** This is the gate. Until K_0 of $\mathbf{Cont}_q$ exists there is no q-analogue of anything in this note, because the minus sign is unavailable.
2. **The q-analogue of Theorem 5.1.** The q-side of F_s should be $\sum z^n/|\mathrm{GL}_n(\mathbb{F}_q)|^s$ — the divisor of the q-Dirichlet series of `q-dirichlet.tex` — and the question is what replaces $\Gamma(\frac{1}{2})$. Jackson's q-integral is the obvious candidate for $\int$, and Γ_q for Γ; whether the self-dual point of Remark 5.2 survives q-deformation is the first thing to check and is cheap.
3. **Prove or refute Conjecture 6.3** on a domain large enough to include a non-hypergeometric example. One such would settle whether the statement is structural or an artefact of the family.
4. **Add the $(\mathbb{N}^\delta, +)$ row** of §2 to `q-dirichlet.tex`, with the observation of Proposition 2.1: its three rows all remember S_n and the fourth does not, which is the distinction the rest of this note is about.

References

[1] François Bergeron, Gilbert Labelle and Pierre Leroux, *Combinatorial Species and Tree-like Structures*, Encyclopedia of Mathematics and its Applications 67, Cambridge University Press, 1998.

[2] Tim Hosgood and David I. Spivak, *Dirichlet polynomials and entropy*, arXiv:2107.04832.

[3] André Joyal, *Une théorie combinatoire des séries formelles*, Adv. Math. **42** (1981), 1–82.

[4] André Joyal, *Foncteurs analytiques et espèces de structures*, in *Combinatoire énumérative*, Lecture Notes in Math. 1234, Springer, 1986, 126–159. Virtual species.

[5] Joachim Kock, *Data types with symmetries and polynomial functors over groupoids*, Proc. MFPS 28, ENTCS 286 (2012), 351–365.

[6] David I. Spivak, *Polynomial functors and Shannon entropy*, arXiv:2201.12878.

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